Load Data from Google Cloud Storage (GCS) to BigQuery Using Airflow (Cloud Composer 3)

This document demonstrates an end-to-end workflow for loading a CSV file from Google Cloud Storage (GCS) into BigQuery using Apache Airflow running on Cloud Composer 3.

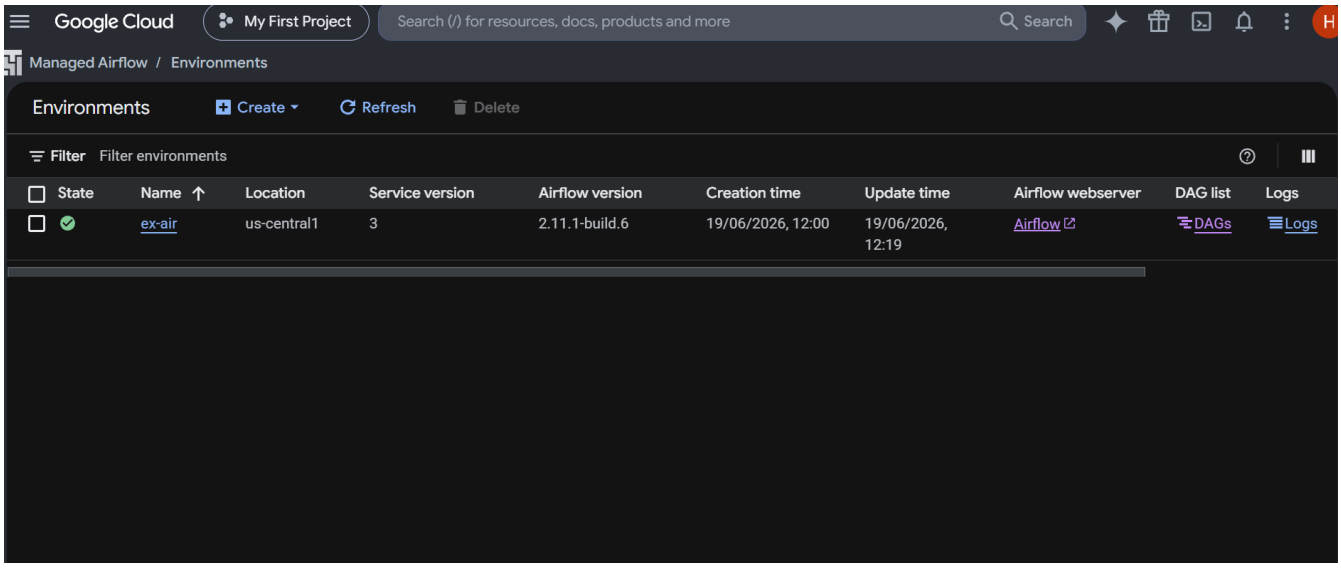
Step 1: Upload CSV File to GCS Bucket

Upload the source CSV file (college_students.csv) into your Cloud Storage bucket.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	student_ic	student_n	gender	departme	year	city	state	admission	fees									
2	1001	Aarav Sha	Male	CSE	1	Hyderaba	Telangana	########	85000									
3	1002	Diya Patel	Female	ECE	2	Vijayawad	Andhra Pr	########	82000									
4	1003	Rohan Gu	Male	MECH	3	Mumbai	Maharash	########	78000									
5	1004	Ananya Re	Female	CSE	4	Bengaluru	Karnataka	########	90000									
6	1005	Arjun Kum	Male	CIVIL	1	Chennai	Tamil Nad	########	75000									
7	1006	Sneha Sin	Female	EEE	2	Pune	Maharash	########	80000									
8	1007	Vikram Ra	Male	CSE	3	Hyderaba	Telangana	########	88000									
9	1008	Priya Nair	Female	ECE	4	Kochi	Kerala	########	81000									
0	1009	Karan Mel	Male	MECH	1	Ahmedab	Gujarat	########	77000									
1	1010	Meera Jos	Female	CSE	2	Jaipur	Rajasthan	########	89000									
2	1011	Rahul Ver	Male	EEE	3	Lucknow	Uttar Prad	########	79000									
3	1012	Kavya Das	Female	CIVIL	4	Bhubanes	Odisha	########	76000									
4	1013	Aditya Mis	Male	CSE	1	Patna	Bihar	########	87000									
5	1014	Nisha Kap	Female	ECE	2	Delhi	Delhi	########	83000									

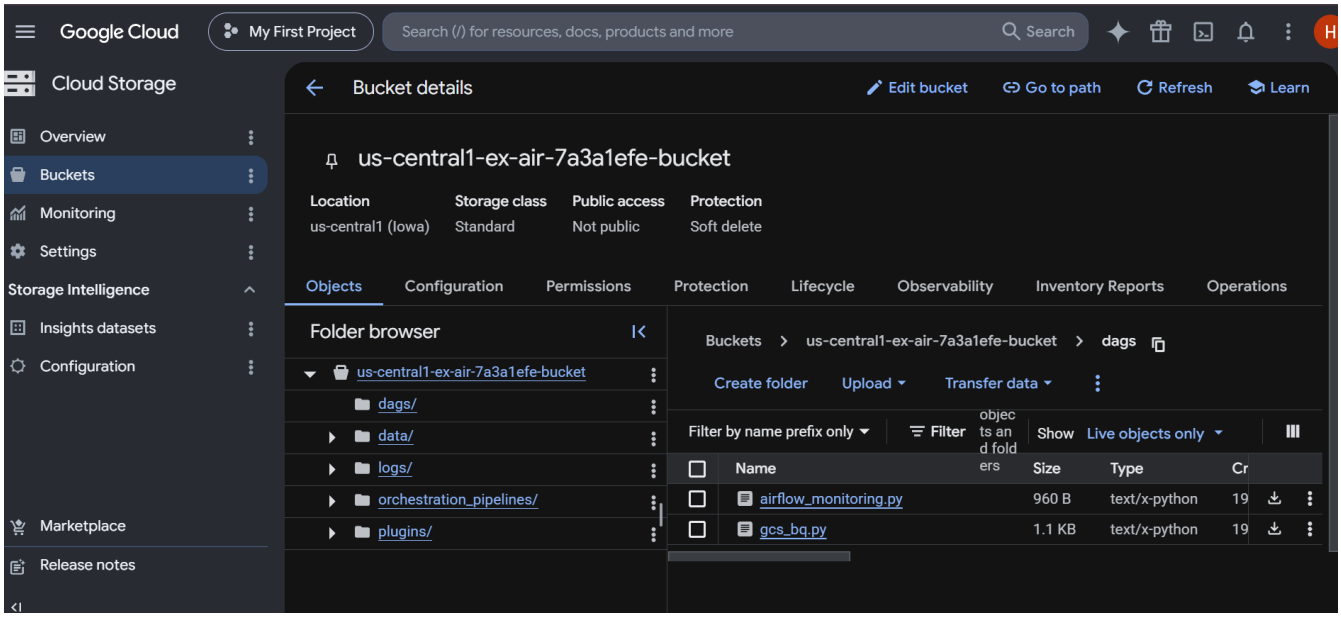
Step 2: Create a Cloud Composer 3 Environment

Create a Composer 3 environment and wait until the environment becomes healthy.



Step 3: Upload DAG File to the Composer DAGs Folder

Upload the DAG Python file into the default Composer bucket under the dags/ folder.



DAG Code

The DAG uses GCSToBigQueryOperator to load CSV data from GCS into a BigQuery table.

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```python
from datetime import datetime

from airflow import DAG
from airflow.operators.empty import EmptyOperator
from airflow.providers.google.cloud.transfers.gcs_to_bigquery import (
    GCSToBigQueryOperator,
)

BUCKET_NAME = "df-demo-007"
FILE_NAME = "college_students.csv"

PROJECT_ID = "project-e1867c45-6603-4a6f-949"
DATASET_ID = "df_exdemo"
TABLE_ID = "students"

with DAG(
    dag_id="gcs_to_bq_two_tasks",
    start_date=datetime(2025, 1, 1),
    schedule=None,
    catchup=False,
) as dag:

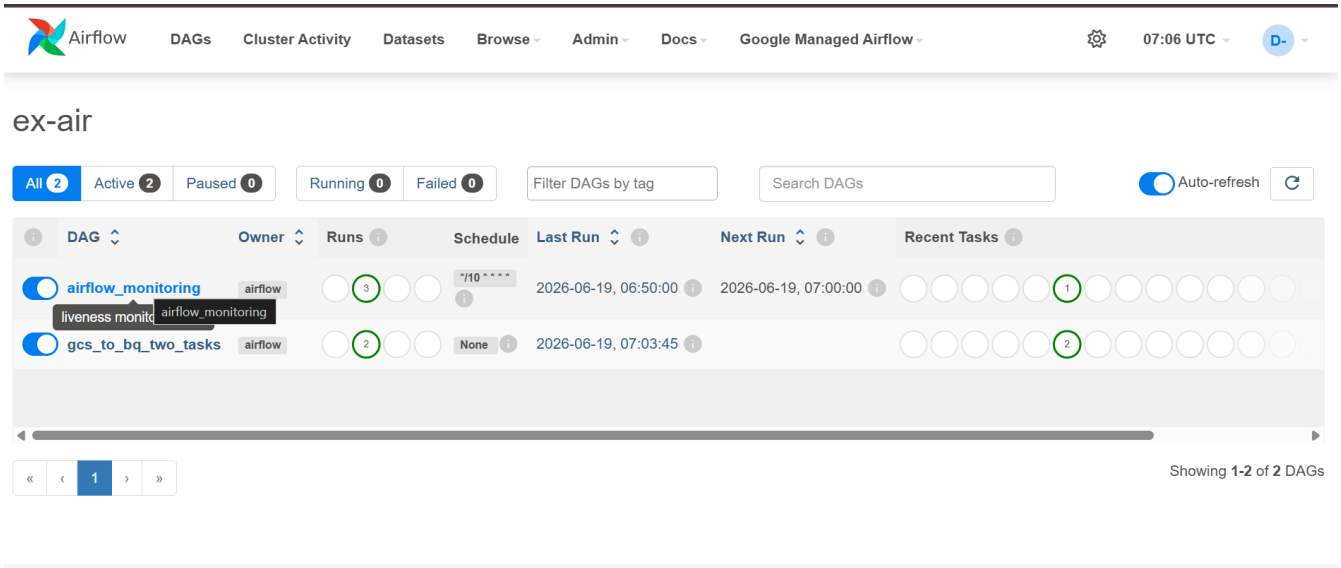
    start = EmptyOperator(
        task_id="start"
    )

    load_to_bq = GCSToBigQueryOperator(
        task_id="load_to_bigquery",
        bucket=BUCKET_NAME,
        source_objects=[FILE_NAME],
        destination_project_dataset_table=(
            f"{PROJECT_ID}.{DATASET_ID}.{TABLE_ID}"
        ),
        source_format="CSV",
        skip_leading_rows=1,
        autodetect=True,
        write_disposition="WRITE_TRUNCATE",
        create_disposition="CREATE_IF_NEEDED",
        gcp_conn_id="google_cloud_default",
    )

    start >> load_to_bq
```

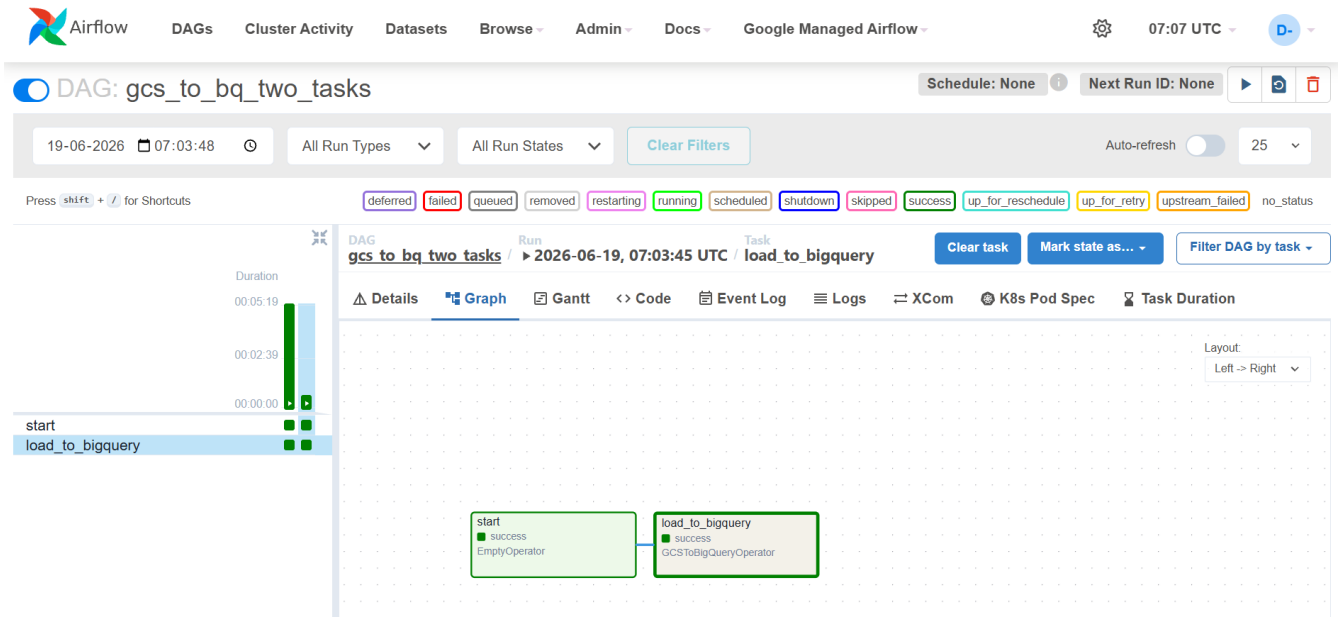
Step 4: Trigger the DAG

Verify that the DAG appears in Airflow and trigger the workflow.



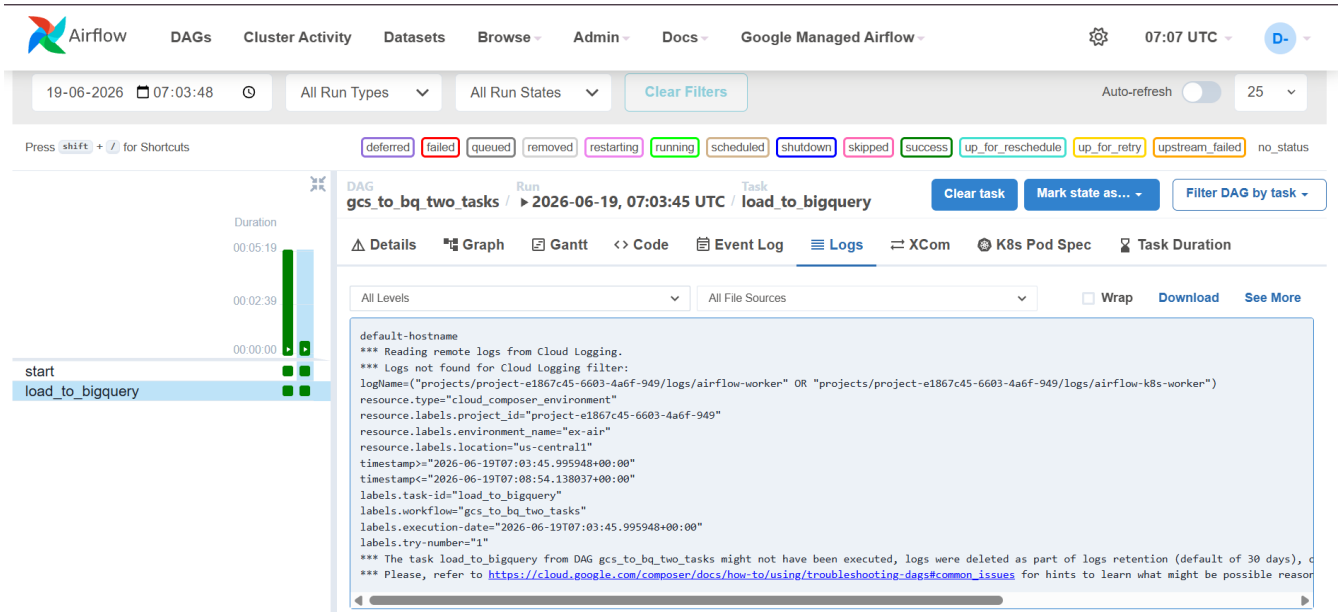
Step 4.1: Verify DAG Graph

Check the Graph view to confirm all tasks completed successfully.



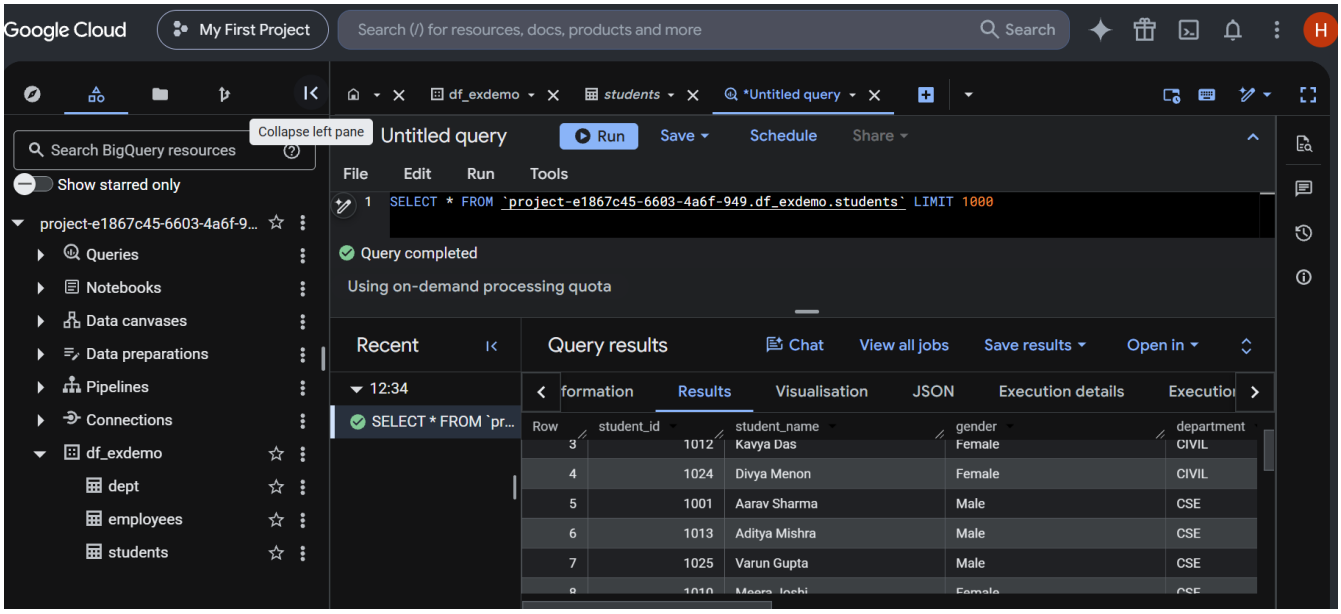
Step 4.2: Check Task Logs

Review Airflow logs for task execution details and troubleshooting.



Step 5: Verify Data in BigQuery

Run a query against the destination table and verify that records were loaded successfully.


Google Cloud
My First Project
Search (/) for resources, docs, products and more
Search
df_exdemo
students
*Untitled query
Collapse left pane
Search BigQuery resources
Show starred only
project-e1867c45-6603-4a6f-9...
Queries
Notebooks
Data canvases
Data preparations
Pipelines
Connections
df_exdemo
dept
employees
students
Untitled query
Run
Save
Schedule
Share
File
Edit
Run
Tools
1 SELECT * FROM `project-e1867c45-6603-4a6f-949.df_exdemo.students` LIMIT 1000
Query completed
Using on-demand processing quota
Recent
12:34
SELECT * FROM `pr...
Query results
Chat
View all jobs
Save results
Open in
formation
Results
Visualisation
JSON
Execution details
Execution
Row
student_id
student_name
gender
department
3 1012 Kavya Das Female CIVIL
4 1024 Divya Menon Female CIVIL
5 1001 Aarav Sharma Male CSE
6 1013 Aditya Mishra Male CSE
7 1025 Varun Gupta Male CSE

Conclusion

The workflow successfully loads data from a CSV file stored in GCS into a BigQuery table using Cloud Composer 3 and Apache Airflow. Validation in BigQuery confirms that the data was loaded correctly.